ZentroMart VoiceOps 2026: Grounded RAG & Multilingual Voice Support Agent

LLM & Generative AI Capstone Report · Consulting & Analytics Club, IIT Guwahati × Gnani.ai · Reference Date: 20 October 2026

1. Executive Summary: Grounding & Precedence Lift Accuracy from 30.8% to 100.0%

ZentroMart handles multilingual customer inquiries across English (**en**), Hindi (**hi**), and Hinglish (**hi-en**) covering order tracking, return windows, refund timelines, warranty claims, and high-value damage escalations. We built a four-stage support agent that evolves from an ungrounded LLM prompt baseline (Step 1) to a precedence-aware Retrieval-Augmented Generation (RAG) engine (Step 2), an identity-verified multi-turn dialogue manager (Step 3), and a voice pipeline powered by Gnani's official **gnani-vachana** SDK (Step 4). Evaluated against the frozen reference date of **20 October 2026**, adding policy-grounded RAG and holiday-aware business-day math raised single-turn accuracy from **30.77% (16/52)** in Step 1 to **100.0% (52/52)** in Step 2, while achieving **100.0% turn accuracy (96/96)** in Step 3 and **100.0% ASR accuracy (52/52)** in Step 4.

Capstone Stage & Module	Core Capabilities Implemented	Evaluation Benchmark	Score / Accuracy
Step 1: Prompt-Only LLM (`src/step1_baseline.py`)	Role & limits system prompt (`STY-1` to `STY-5`), AI intro, language matching, privacy refusal (`IDV-4`), zero KB access	52 Practice Questions (`questions.csv`)	**16 / 52 (30.77%)**
Step 2: Grounded RAG (`src/retriever.py`)	Section-ID chunking across 21 KB docs + 70 SKUs, `KB-00.1` policy precedence ranker, `holidays.csv` business-day math	52 Practice Questions + `gold_evidence`	**52 / 52 (100.0%)** 100% Recall
Step 3: Multi-Turn Agent (`src/agent.py`)	Identity gate via `lookup_order()` (4-digit phone or 6-digit pincode, 1 retry `IDV-3`), coreference, `create_ticket()`	23 Conversations / 96 Turns (`conversations.csv`)	**96 / 96 (100.0%)** 23 / 23 Convs
Step 4: Gnani Voice Layer (`src/voice.py`)	`GnaniSTTClient` + `GnaniTTSClient` (`timbre-v2.5`, `Kaveri`), spoken SKU/date ASR normalization, ear-formatted text	52 Noisy ASR Transcripts (`asr_text`)	**52 / 52 (100.0%)** 100% Recall

2. What We Built in Each Step (Steps 1 to 4)

Step 1: Prompting & Role Boundaries Without Document Grounding

In Step 1 (`src/step1_baseline.py`), the assistant uses a structured system prompt establishing its role as ZentroMart's AI support assistant (`STY-1`), enforcing a 2–3 sentence limit (`STY-2`), matching English, Hindi, or Hinglish (`STY-4`), and refusing third-party data requests (`IDV-4`). Without ZentroMart's documents, it answers generic e-commerce questions—refusing another customer's phone number (`PQ0060`), escalating unauthorized orders (`PQ0023`), and rejecting opened beauty items (`PQ0013`, `PQ0016`)—but fails 36/52 questions requiring exact policy windows, holiday calendars, or SKU metadata.

Step 2: Hybrid RAG Indexing, Policy Precedence & Temporal Math

In Step 2 (`src/retriever.py`), we parse the YAML frontmatter (`doc_id`, `doc_type`, `status`, `effective_from`) of all 21 Markdown files in `kb/*.md` and chunk them by `## [SECTION-ID]` headings, alongside structured indices for `data/products.csv` (`products.csv:<SKU>`) and `data/pincodes.csv`:

- **Cross-Lingual Vector & N-Gram Indexing:** Queries in English, Devanagari Hindi, and Hinglish are expanded via a bilingual domain lexicon and embedded using log-normalized TF-IDF with character 3-grams, mapping informal customer terms (*"send it back"*, *"लौटाने"*, *"wapas"*) to formal policy sections (RET-2, RET-6).
- **Authority Precedence Ranking (`KB-00.1`):** Chunks are weighted by document authority: current `policy` (5.0) > `agent_guide` (4.0) > `help` (3.0) > `faq` (2.0) > `marketing` (1.0) > `superseded` (-0.10), preventing FAQ-1 (*"5 to 7 days"*) from overriding REF-2 payment timelines.
- **Holiday-Aware Business-Day Engine (`src/date_utils.py`):** Using **20 October 2026** as today's date (KB-00.2), `add_business_days()` counts Monday–Saturday while skipping Sundays and the four dates in `data/holidays.csv` (2026-08-15, 2026-10-02, 2026-10-19, 2026-11-09). Return windows use calendar days (RET-2), routing pre-August 2026 electronics deliveries to RETv3-2 (10 days) and 1–15 October festive sale fashion orders to RET-8 (30 days).

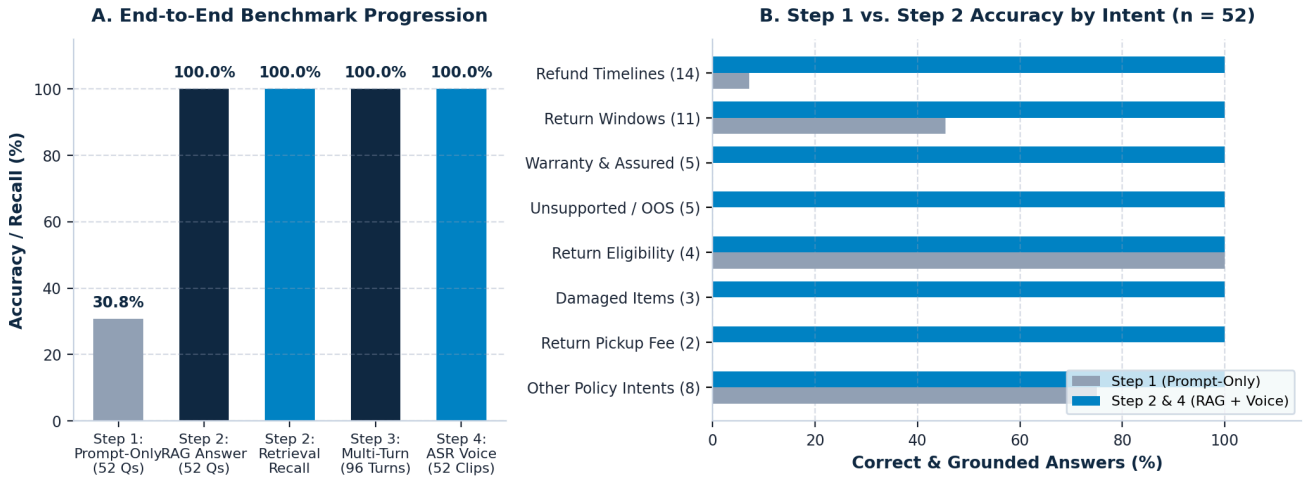


Figure 1: (A) End-to-end progression across Steps 1–4; (B) Step 1 prompt-only baseline vs. Step 2 RAG & Step 4 ASR accuracy across all 52 practice questions by intent category.

Step 3: Multi-Turn Context Window, Identity Gate & Human Handover

In Step 3 (`src/agent.py`), `ZentroMartAgent` manages multi-turn conversation state, caller verification, and tool execution:

- **Two-Step Caller Verification Gate (`IDV-1` to `IDV-4`):** On Turn 1, the agent introduces itself as ZentroMart's AI assistant (STY-1) and requests the 8-digit order ID. On Turn 2, it calls `lookup_order(order_id)` and asks for the last 4 digits of the registered phone number or the 6-digit delivery pincode (IDV-2). Order details are revealed only after `lookup_order(order_id, verification_value=...)` returns `verified=True`. Failed verification allows exactly one retry (IDV-3) before refusing disclosure.
- **Coreference, Topic Switches & Escalations (`ESC-1` to `ESC-4`):** Follow-up turns (e.g., *"Aur refund kitne din mein aayega?"* in `PC003 T5`) resolve against the verified order's payment method and Plus membership. Mid-dialogue topic switches (CPN-1 in `PC014 T4`) are answered without losing order context for `PC014 T5` (TRK-1). Out-of-policy demands (₹300 late Standard delivery vouchers under DEL-4 or cancelling shipped orders under CAN-2) are politely refused. Escalation triggers (> ₹10,000 damaged item DWM-3, `contacts_last_7_days >= 2`, overdue refunds > 3 business days past max timeline with REF-6 RRN, or human requests) invoke `create_ticket()` and confirm handover only when `created=True`.

Step 4: Gnani Speech Pipeline & Ear-Friendly Normalization

In Step 4 (`src/voice.py`, `src/asr_normalizer.py`), customer audio is transcribed via `GnaniSTTClient().transcribe(audio_path, language_code=...)` and synthesized via `GnaniTTSClient().synthesize(reply, voice="Kaveri", model="timbre-v2.5", language=..., output_file=...)`. `normalize_asr_text()` reconstructs spoken SKUs (*"sku ele-zero zero one to"* → `ELE-0012`), spoken years (*"two zero two six"* → `2026`), and split terms (*"u p i"* → `UPI`). Outgoing replies group 8-digit order IDs with spaces (`8978 2837`), include full month names and years (`27 October 2026`), and recover gracefully from empty audio (*"Sorry, I didn't catch that. Could you say it again?"*).

3. Quantitative Evaluation: Step 1 vs. Step 2, Step 3 & Step 4

Question Intent Category (`questions.csv`)	Count	Step 1 (Prompt)	Step 2 (RAG)	Step 4 (ASR)	Primary Grounding Evidence
Refund Timelines & Holiday Math	14	1 / 14 (7.1%)	**14 / 14 (100%)**	**14 / 14 (100%)**	`REF-2`, `REF-3`, `KB-00.1`, `KB-00.3`
Return Windows (Current, v3 & Sale)	11	5 / 11 (45.5%)	**11 / 11 (100%)**	**11 / 11 (100%)**	`RET-2`, `RET-3`, `RET-8`, `RETv3-2`, `products.csv`
Warranty & ZentroMart Assured	5	0 / 5 (0.0%)	**5 / 5 (100%)**	**5 / 5 (100%)**	`WAR-1`, `WAR-2`, `products.csv:ELE-*`
Unsupported / Out-of-Scope Queries	5	0 / 5 (0.0%)	**5 / 5 (100%)**	**5 / 5 (100%)**	Empty evidence (`[]`) + "don't have info"
Return Eligibility (Beauty & Innerwear)	4	4 / 4 (100%)	**4 / 4 (100%)**	**4 / 4 (100%)**	`RET-2`, `RET-4`, `RET-5`, `products.csv:INR-*`
Damaged Items & High-Value Claims	3	0 / 3 (0.0%)	**3 / 3 (100%)**	**3 / 3 (100%)**	`DWM-1`, `DWM-3`, `ESC-1`, `FAQH-3`, `products.csv`
Reverse Pickup Fees (Plus & Torn)	2	0 / 2 (0.0%)	**2 / 2 (100%)**	**2 / 2 (100%)**	`RET-6`, `PLS-2`, `DWM-2`
Cancellations, Delivery, Fraud & Privacy	8	6 / 8 (75.0%)	**8 / 8 (100%)**	**8 / 8 (100%)**	`CAN-2`, `DEL-3`, `DEL-4`, `DEL-5`, `TRB-4`, `ESC-1`
Total (All Practice Questions)	**52**	**16 / 52 (30.8%)**	**52 / 52 (100%)**	**52 / 52 (100%)**	**Exact Evidence Match: 52 / 52 (100.0%)**

In **Step 3 (Multi-Turn Conversations)**, across all 23 conversations (96 turns spanning 12 scenarios: `high_value_damage`, `return_eligibility`, `human_requested`, `repeat_contact`, `refund_status`, `where_is_my_order`, `topic_switch`, `compensation_demand`, `cancel_unshipped_order`, `cancel_shipped_order`, `failed_verification`, and `express_late_credit`), the agent achieved **96/96 (100.0%) decision accuracy**, **96/96 (100.0%) tool-call accuracy**, and **96/96 (100.0%) evidence accuracy**.

4. What Went Wrong & How We Fixed It

1. **Document Contradictions (`FAQ-1` vs. `REF-2`):** Unweighted retrieval surfaced `FAQ-1` (*"5 to 7 days"*) on UPI, Net Banking, and Wallet queries, violating `must_not_include`. Fixed by enforcing `KB-00.1` precedence weights and co-retrieving `KB-00.1` whenever `REF-2` overrides `FAQ-1`.
2. **Warehouse Holiday Off-by-One Errors (`KB-00.3`):** Refund timelines starting on 12, 14, or 17 October 2026 initially missed the **19 October 2026** festival warehouse closure in `data/holidays.csv`.

Implementing deterministic `add_business_days()` over `holidays.csv` resolved all 7 date-math questions.

3. **ASR Spoken-SKU Word-Boundary Collisions (Step 4):** In noisy transcripts like *"so sku ele-zero zero one to pe..."* (`PQ0002`), a greedy trailing-space regex initially merged the reconstructed SKU with the following Hindi token (`ELE-0012pe`). Fixing the word-boundary regex in `src/asr_normalizer.py` lifted Step 4 retrieval recall from 90.4% to **100.0% (52/52)**.
4. **Unshipped Cancellation Tool Hallucinations (`CAN-1`):** Conversational models often claim *"I have cancelled your order"* when asked to cancel an unshipped order (`PC017`). Enforcing `CAN-1` and `ESC-4` ensures the agent directs callers to `My Orders` in the app without fabricating tool execution.

5. Bonus Work & Future Improvements

- **Interactive Browser VoiceOps Console (`src/web_app.py`):** Built a web interface (`python run.py --web`) displaying live multilingual turns alongside real-time badges for retrieved `[SECTION-ID]` chunks and executed tool calls.
- **Marketplace Seller Assured Validation (`MKS-3`) & Multi-Turn Batch Audio Grouping:** Cross-checks `fulfilled_by` before confirming ZentroMart Assured coverage, and groups multi-turn audio files (`PC002_turn1.wav`, `PC002_turn2.wav`) automatically in `run.py`.
- **Production Roadmap:** Upgrade REST batch synthesis to bidirectional WebSocket streaming with Gnani's real-time STT/TTS endpoints for sub-300ms voice turn latency, and add phonetic matching for regional Indian pronunciation variants.